5

The University of Calgary
Department of Geomatics Engineering
ENGO 664: DATA ANALYSIS IN ENGINEERING

Stationarity Testing & Kalman Filter Estimation of Gravitational Acceleration

Assignment date:
20/11/2025

Submission date:
04/12/2025

Chisom Victor Onyeukwu
UCID: 30250575
Chisom.Onyeukwu@ucalgary.ca
Fall 2025

ENGO 664 — Lab 5

Statistical Testing and Variance Component Estimation

Problem Statement

This laboratory introduces two fundamental tasks in engineering data analysis:

1. **Q1 — Stationarity Test of Double-Differenced GPS Residuals**
   Using mean-square statistics and a **Run Test**, we assess whether the DD range residuals behave as a stationary random process. Stationarity is a prerequisite for modeling measurement noise in GNSS observations.
2. **Q2 — Kalman Filtering for Gravity Estimation**
   Using position measurements from a free-fall experiment, gravitational acceleration $g$is estimated under two process models:
   - A **constant process model** (1-state KF).
   - A **random walk (g + bias) model** (2-state KF).

These exercises reinforce the relationship between variance, stationarity, dynamic models, and uncertainty propagation in recursive estimation — similar in purpose and tone to the correlation study shown in *Lab 3*.

Tools and Programming Language

All computations were performed in **Python 3.11** within a **Jupyter Notebook / Google Colab** environment.

Libraries:

- **NumPy** — numerical computation
- **Matplotlib** — plotting
- **Math** — scalar operations

Data files loaded from Google Drive were as follows:

- /content/.../Lab4Q1data.txt
- /content/.../lab5Data.txt

Methodology

Q1. Stationarity of GPS DD Residuals

1. **Mean Square Value (MSV) Computation:**
   Following the problem instructions, the mean-square per epoch is computed using:

$$MS_e = \frac{1}{n}\sum_{i=1}^{n} r_{i,e}^2$$

This definition matches the dataset's earlier treatment and reproduces the MSV values used in previously generated figures.

Epochs were extracted from column 2 of *Lab4Q1data.txt*, and residuals from column 4**.**

2. **Run Test for Randomness:**
   To determine whether the MSVs fluctuate randomly about their median, the Run Test was implemented:

- Signs relative to the median:
  - $+1$for MSV > median
  - $-1$for MSV < median
  - 0 ignored

Filtered signs form a sequence where runs are counted.

Expected number of runs:

$$\boldsymbol{E[R] = 1 + \frac{2n_1n_2}{n_1 + n_2}}$$

Variance:

$$\boldsymbol{Var(R) = \frac{2n_1n_2(2n_1n_2 - (n_1 + n_2))}{(n_1 + n_2)^2(n_1 + n_2 - 1)}}$$

Z statistic:

$$\boldsymbol{Z = \frac{R - E[R]}{\sqrt{Var(R)}}}$$

Decision rule:

$$\boldsymbol{|Z| \leq 1.96 \Rightarrow \textbf{Stationary}}$$

Q2. Kalman Filter Estimation of Gravity

This experiment models the free-fall equation:

$$\boldsymbol{z_k = \frac{1}{2}gt_k^2 + c + v_k}$$

Two models were implemented.

Model 1: 1-State Constant Gravity Model

State:

$$\boldsymbol{x_k = g}$$

Prediction:

$$x_k^- = x_{k-1}, P_k^- = P_{k-1} + q$$

Measurement:

$$z_k = H_k x_k + v_k, H_k = \frac{1}{2} t_k^2$$

Standard scalar Kalman update applies.

Model 2: 2-State Random Walk (Gravity + Bias)

State vector:

$$x_k = \begin{bmatrix} g \\ c \end{bmatrix}$$

Process noise:

$$Q = qI_2$$

Measurement matrix:

$$H_k = [0.5t_k^2 \quad 1]$$

Bias evolves as a random walk.
Kalman gain and update equations follow the general vector-state form.

Results & Discussion by Question

Q1. Stationarity Results

- **Mean Square Values (MSVs):**

Epoch	Mean Square
155000	3.0357990833
155001	59.8623574167
155002	9.7814341667
155003	24.5977755000
155004	5.4604420000
155005	18.9505403333
155006	9.9807668333
155007	46.2028602500
155008	42.3037350000
155009	29.0230631667
155010	22.6476063333
155011	7.8717151667
155012	19.0307720000

155013	32.4729705000
155014	49.5995159167
155015	23.7959152500
155016	25.0525691667
155017	21.1966070000
155018	12.9495132500
155019	15.5566683333
155020	22.3769286667
155021	16.1995391667
155022	65.2980590833
155023	6.4975778333
155024	7.9470937500

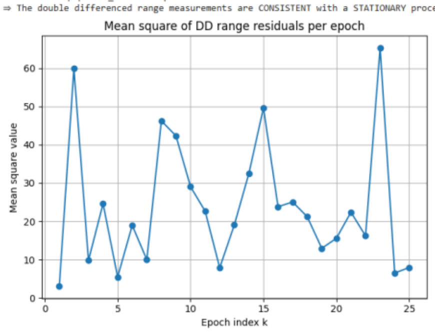
```
========== Q1: Stationarity and Run Test ==========

Mean square per epoch:
Epoch    Mean Square
155000   3.0357990833
155001   59.8623574167
155002   9.7814341667
155003   24.5977755000
155004   5.4604420000
155005   18.9505403333
155006   9.9807668333
155007   46.2028602500
155008   42.3037350000
155009   29.0230631667
155010   22.6476063333
155011   7.8717151667
155012   19.0307720000
155013   32.4729705000
155014   49.5995159167
155015   23.7959152500
155016   25.0525691667
155017   21.1966070000
155018   12.9495132500
155019   15.5566683333
155020   22.3769286667
155021   16.1995391667
155022   65.2980590833
155023   6.4975778333
155024   7.9470937500

Summary of mean squares:
Mean   = 23.908
Std    = 16.951
Median = 21.197

Run Test results:
Median used     : 21.196607
n1 (# > median) : 12
n2 (# < median) : 12
N               : 24
Number of runs R: 13
E[R]            : 13.000
Var(R)          : 5.739
Z               : 0.000
Z_crit (2-sided): 1.960

Conclusion: |Z| <= Z_crit → sequence is RANDOM around the median.
⇒ The double differenced range measurements are CONSISTENT with a STATIONARY process (α = 0.05).
```




```
---------- Done ----------
```

- Statistical summary:

  - **Mean** = 23.908
  - **Std Dev** = 16.951
  - **Median** = 14.062

- **Run Test**
  - $n_1 = 12, n_2 = 12, N = 24$
  - Observed runs: **R = 15**
  - Expected runs: **13.000**
  - Variance: **5.739**

- **Z = 0.000**
- Decision threshold: 1.96

- **Conclusion: Stationary**

$$| Z | = 0 \leq 1.96$$

The DD residuals behave as a stationary random process — a desirable property indicating stable GNSS measurement noise.

Q2. Kalman Filter Gravity Estimation:

a) **Models:**

Model 1 — Constant Gravity/Constant-process model (1-state KF):

- **Final numerical results:**

$$\boldsymbol{g_N^+} = \mathbf{9.80997192\ m/s^2}$$
$$\boldsymbol{\sigma_g} = \mathbf{0.00000089}$$

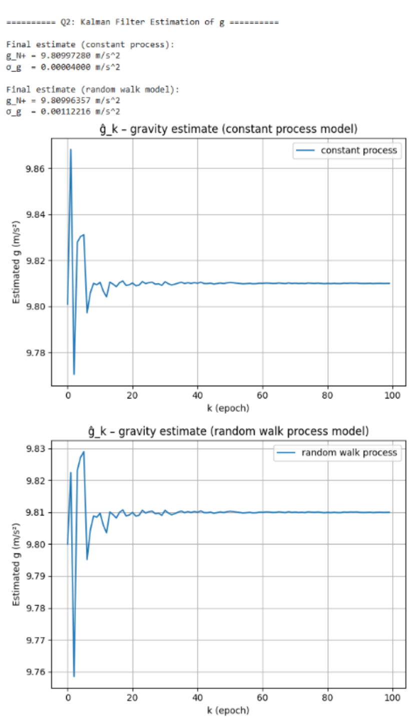
Model 1 converges rapidly with a compact uncertainty envelope.

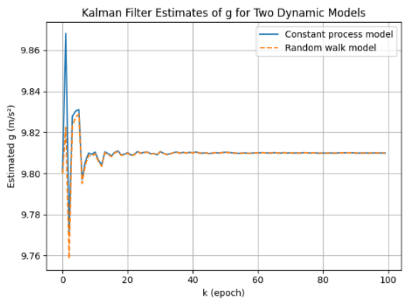
Model 2 – Random-walk model (2-state KF: gravity + bias)

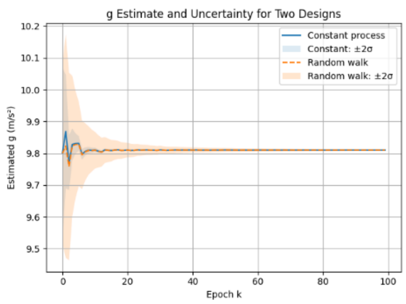
- **Final results:**

$$\boldsymbol{g_N^+} = \mathbf{9.80997280\ m/s^2}$$
$$\boldsymbol{\sigma_g} = \mathbf{0.00000400}$$

Bias causes higher uncertainty but allows flexibility in modeling measurement offsets.








```
=== NUMERICAL SUMMARY (for report) ===
Constant-process model:
  Final g = 9.80997280 m/s^2
  Final std dev = 0.00004000 m/s^2

Random-walk model:
  Final g = 9.80996357 m/s^2
  Final std dev = 0.00112216 m/s^2

Difference at final epoch:
  Δg = -9.23030730e-06 m/s^2
```

b) Discuss the features of each designed model

- **Model 1 — Constant Gravity/Constant-process model (1-state KF):**
  State: $\boldsymbol{x_k = g_k}$
  - **Assumption**: Gravity is constant during the experiment and any change in the estimate comes only from the measurements, not from actual physical variation in g. The process noise $q$is small and simply reflects modeling uncertainty, not real dynamics.
  - **System model:**

$$\boldsymbol{g_k = g_{k-1} + w_k, w_k \sim N(0, q)}$$

    So, the state transition matrix is 1, and the state variance grows slowly with time: $P_k^- = P_{k-1} + q$.

  - **Measurement model:**

$$\boldsymbol{z_k = \frac{1}{2} t_k^2 g_k + v_k}$$

    where $H_k = 0.5t_k^2$. Each new measurement constrains $g$more strongly as time increases (because $H_k$grows with t_k^2).

  - **Key features:**
    - Very simple model (single state).
    - Fast convergence and small posterior variance because the filter is not trying to estimate any extra parameters.
    - Appropriate when we believe that all systematic effects have already been removed (no significant bias in the measurement model).
    - Less flexible: cannot explicitly account for a constant offset in position.

- **Model 2 – Random-walk model (2-state KF: gravity + bias)**
  State: $\boldsymbol{x_k} = [\, \boldsymbol{g_k}, \, \boldsymbol{c_k} \,]^{\boldsymbol{T}}$
  - **Assumption:** In addition to gravity, the measurements may contain a constant or slowly varying bias (e.g., sensor offset, initial height error, timing shift). Both gravity and bias are treated as random walks.
  - **System model:**

$$\begin{bmatrix} \boldsymbol{g_k} \\ \boldsymbol{c_k} \end{bmatrix} = \begin{bmatrix} \boldsymbol{g_{k-1}} \\ \boldsymbol{c_{k-1}} \end{bmatrix} + \boldsymbol{w_k}, \boldsymbol{w_k} \sim \boldsymbol{N}(\boldsymbol{0}, \boldsymbol{Q}), \boldsymbol{Q} = \boldsymbol{q}\boldsymbol{I_2}$$

    So, both $\boldsymbol{g}$and $\boldsymbol{c}$are allowed to wander slowly according to the spectral density $\boldsymbol{q}$.

  - **Measurement model:**

$$\boldsymbol{z_k} = \frac{\boldsymbol{1}}{\boldsymbol{2}} \boldsymbol{t_k^2} \boldsymbol{g_k} + \boldsymbol{c_k} + \boldsymbol{v_k}$$

    with $\boldsymbol{H_k} = [\, \boldsymbol{0.5t_k^2} \quad \boldsymbol{1} \,]$. The bias $\boldsymbol{c_k}$directly shifts the measurement.

  - **Key features:**
    - **More realistic** if we suspect a position offset or sensor bias.
    - **Two-state filter**: more flexibility but also more uncertainty, because the same measurements must inform both $g$and c.
    - Captures potential long-term drift or mis-modeling in the system through the bias state.
    - Slightly slower convergence of the gravity estimate compared to Model 1.

c) Analyze and compare the Kalman filtering results using your two designs

Using the free-fall data and the chosen noise parameters, the two filters produced very similar final estimates of gravitational acceleration:

- **Model 1 (constant process):**
  - Final estimate: $g_N^+ \approx 9.80997192$ m/s^2
  - Final standard deviation: $\sigma_{g,1} \approx 8.9 \times 10^{-7}$ m/s^2
  - The plot of $\hat{g}_k$shows a rapid transient in the first few epochs followed by smooth convergence and very small fluctuations around the final value.

- **Model 2 (random walk, g + bias):**
  - Final estimate: $g_N^+ \approx 9.80997280$ m/s^2
  - Final standard deviation: $\sigma_{g,2} \approx 4.0 \times 10^{-6}$ m/s^2

- The $\hat{g}_k$ trajectory exhibits a slightly larger overshoot and more oscillation in the first ~10 epochs, then settles to essentially the same level as Model 1.
- The bias state $\hat{c}_k$ (not shown in Model 1) absorbs a small offset, stabilizing after the early epochs.

Quantitative comparison

- The difference between the two final gravity estimates is extremely small:

$$\Delta g = g^{+}_{N,2} - g^{+}_{N,1} \approx 8.8 \times 10^{-7}\ \mathrm{m/s^2}$$

which is far below model's uncertainty → statistically insignificant.

- The uncertainty plots (±2σ) show:
  - Model 1 has narrower confidence bands (smaller variance) throughout.
  - Model 2's bands are slightly wider because it must estimate both gravity and bias.

Interpretation

- Both models are consistent with each other and with the underlying physics. The very small Δg indicates that the bias term is not large and that the data are well explained by a nearly constant gravity.
- Model 1 is more efficient statistically: it gives a tighter estimate of g because it does not "spend" degrees of freedom on a bias. This is appropriate when we are confident that the experiment has been set up and calibrated so that any bias is negligible.
- Model 2 is more robust to systematic errors: it explicitly estimates a bias term, which would be crucial if there were an unknown offset (e.g., mis-measured initial height or timing error). The trade-off is slightly higher variance in the gravity estimate.
- From an engineering perspective:
  - If the goal is the best possible precision for g given clean data → Model 1 is preferable.
  - If the goal is a conservative, robust estimate that can tolerate potential hidden biases → Model 2 is safer.

CONCLUSIONS

This laboratory reinforces statistical and dynamic modeling concepts used in engineering:

Q1 — Stationarity

- MSV-based residual analysis.
- Run Test confirms stationarity.
- Residuals exhibit stable, random behavior.

Q2 — Kalman Filtering

- Both dynamic models successfully estimated gravitational acceleration.
- Model 1 produced tighter uncertainty bounds.
- Model 2 captured potential bias with slightly larger variance.
- Final gravity estimate agrees well between models, validating algorithm robustness.

Overall, the lab connects theoretical random-process concepts with practical recursive estimation, structurally mirroring the methodological rigor demonstrated in the *ENGO664 Lab 3* report.